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MILITARY SPECIFICATION

GYROSCOPE, RATE, SWITCHING, TYPE MC-1

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of rate switching gyroscope, designated Type MC-1.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-83488	Coating, Aluminum, Ion Vapor Deposited

"Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter."

FSC 6615

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STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Verification/Demonstration/ Evaluation
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-882	System Safety Program for System and Subsystems and Equipment: Requirement for
MIL-STD-889	Dissimilar Metals
MS3106	Connector, Plug, Electric, Straight

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4 and 6.5).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials and electronic parts

3.3.1 Materials.

3.3.1.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the gyroscope, except where magnetic materials are essential.

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3.3.1.2 Protective treatment. When materials are used in the contruction of the gyroscope that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided. Components contained within hermetically sealed containers shall be considered to be adequately protected against deterioration.

3.3.1.3 Fungusproof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are used in a hermetically sealed enclosure fungicidal treatment will not be necessary.

3.3.1.4 Corrosive fumes. The materials, as installed in the gyroscope and under the service conditions specified herein, shall not liberate deleterious fumes.

3.3.1.5 Metals. Metals shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt fog, or atmospheric conditions likely to be met during storage or normal service.

3.3.1.5.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.1.6 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible without jeopardizing the intended end use of the item.

3.3.2 Electronic parts. Electronic parts and the application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on aproved lists shall not be used unless approved by the procuring activity.

3.4 Design and construction. The gyroscope shall be designed to provide switching during turns, and shall be sufficiently rugged to meet the requirements specified herein. The gyroscope shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jar, vibrations, and other conditions incident to shipping, storage, installation, and service.

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3.4.1 Safety. Hazards associated with use of this equipment shall be controlled as specified in MIL-STD-882, Production Phase.

3.4.2 Reliability. The gyroscope shall have a minimum acceptable mean-time-between-failures (MTBF) of 2,000 hours based upon a discrimination ratio of 1.5 and a confidence factor of 90 percent. (See 6.4.1 and 6.4.2.)

3.4.2.1 Longevity. The gyroscope shall have an operating life span of not less than 3,000 hours before the equipment consistently fails to meet the specified MTBF index.

3.4.3 Maintainability. The gyroscope shall be so constructed that adjustments and repairs can be easily made by the personnel of overhaul bases without the use of special tools unless they are approved by the procuring activity. The following maintainability requirements shall be considered in the design of the gyroscope:

a. Minimization of the complexity of maintenance tasks (for example· calibration, adjustments, inspection, etc) by maximum use of simple design which includes optimum interchangeability and use of standardized equipment or commercial items

b. Rapid and positive recognition of equipment malfunction or marginal performance

c. Rapid and positive identification of the replaceable defective part, assembly or component

d. Minimum numbers and types of tests and tests equipment (special and standard) needed to perform maintenance.

3.4.4 Power. The gyroscope shall operate from 115-V, single-phase, 400 Hz power. There shall be no internal grounding within the instrument. The gyroscope shall function satisfactorily with variations in voltage and frequency ranging from 103 to 127V and 380 to 420 Hz, respectively.

3.4.5 Electrical receptacle. A hermetically sealed receptacle shall be affixed to the case. The receptacle shall accept or mate with an MS3106E14S-5S plug, or equivalent. The electrical receptacle shall be equipped with a water-vaporproof cap and gasket. The pin connections shall be as follows:

Pin		Circuit
A	115V, 400 Hz	Single-phase a-c power
B	115V, 400 Hz	
C	Turn switching contact terminal (normally closed)	

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D Turn switching contact terminal (common)

E Turn switching contact terminal (normally open)

3.4.6 Switching mechanism. The gyroscope shall incorporate a rate gyro of a design suitable to provide the switching characteristics of the outputs as specified herein. The outputs shall be connected to pins C, D, and E of the electrical connector. The circuit between pins D and E shall be normally open and the circuit between C and D shall be normally closed when power is disconnected or when there is zero rate of turn imposed. When a rate of turn of .15° per minute or more is imposed on the gyroscope, the circuit between pins C and D shall open and the circuit between D and E shall close. The outputs shall return to normal within 15 seconds after this turn rate has ceased. The output circuits shall be isolated from the other circuitry in the gyroscope and from the case.

3.4.6.1 Output compatibility. When either output circuit is closed, the maximum resistance between pins shall be 2 ohms. When either output circuit is open, the minimum impedance at 400 Hz shall be 250K. Either circuit shall be capable of switching 750 milliampere at ±200V.

3.4.7 Spin motor. The spin motor shall not incorporate slip rings, outputs, brushes, or any similar element that may fail due to wear incurred in normal operation. During assembly, the rotors shall be readjusted after the 60-hour running time.

3.4.8 Stops. Stops shall be mounted in the gyroscope to prevent damage in extreme positions.

3.4.9 Damping. The gyroscope shall be of a symmetrical nongaseous fluid-damped design and shall be temperature-conpensated without the use of heaters.

3.4.10 Rubbing parts. The gyroscope shall not incorporate rubbing parts of components which would be subject to wear, except for bearings.

3.4.11 Case. The gyroscope case shall be made of sturdy material which shall be uniform in texture and shall be covered in durable lusterless black. It shall be hermetically sealed (see 6.4.3) and of such design that the internal mechanism may be removed from the case as a unit, replaced, and the case resealed without the use of special tools unless they are approved by the procuring activity. The seal shall be accomplished in such a manner that it will not be dependent upon materials that might be affected by the action of any atmosphere to which the gyroscope may be subjected. The case shall be so rugged that it will not become distorted to such extent as to degrade the performance of

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the gyroscope. The case shall be marked to indicate the location of the sensitive axis for proper installation in the aircraft.

3.4.11.1 Filling medium. The filling medium in the case shall be at least 98 percent purity, and shall contain not more that 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi. If the gyroscope is a floated unit and is sealed in a compartment of the case separate from other components, this requirement shall apply only to the compartment containing the remaining components.

3.4.12 Mounting orientation. The gyroscope shall be so mounted that the plane of the mounting feet will be perpendicular to the turn-rate sensitive axis. The gyroscope shall operate properly in the normal position with its feet down and in the inverted position.

3.4.13 Shock mount. Shock mounts shall not be used between the gyroscope and the airframe.

3.4.14 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.4.15 Soldering. Soldering shall be accomplished in accordance with requirement 5 of MIL-STD-454.

3.5 Performance. The gyroscope shall be capable of satisfactory operation when subjected to any combination of the following conditions:

a. Temperatures - ambient operating temperatures ranging from -54° to +74°C and operation for 10 minutes at +98°C, and storage temperatures ranging from -62° to +95°C

b. Humidity - relative humidity up to 100 percent

c. Vibration

(1) Error - vibration at a double amplitude of 0.060 inch through a frequency range of 10 to 55 Hz

(2) Failure - acceleration of 5g through a frequency range of 100 to 500 Hz and 1g from 5 to 23 Hz

d. Acceleration - acceleration forces up to 20g

e. Altitude - pressures ranging from 30 inches Hg (sea level) down to 0.082 inch Hg (approximately 80,000 feet)

f. Fungus - exposure to fungus enzymes

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g. Salt-fog - exposure to salt-fog for a minimum period of 168 hours

h. Temperature shock - temperature shock caused by an abrupt temperature drop from +71° to -54°C

i. Dynamic shock - dynamic shocks at accelerations up to 20g for a nominal duration of 10 milliseconds.

j. Exposure to acoustic sound fields that are characteristic of aircraft.

3.5.1 Electromagnetic interference and susceptibility. The gyroscope shall meet the electromagnetic interference and susceptibility requirements of MIL-STD-461 as specified for class A1 equipment, except that the short-duration interference requirement shall not apply.

3.5.2 Thermal shock. The gyroscope case shall show no evidence of leakage when subjected to alternate immersions in water at a temperature of 85°C and then in water at a temperature of 5°C.

3.5.3 Leak rate. The gyroscope shall be so sealed that the maximum leak rate will not permit more than 10 percent contamination of the filling medium after 1,000 hours at a pressure differential equivalent to 1 atmosphere.

3.5.4 Power consumption. The total power consumption of the gyroscope shall not exceed 10 va while operating nor 14 va while starting.

3.5.5 Undervoltage. The gyroscope shall not be damaged by voltage below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.5.6 Dielectric strength. There shall be no breakdown of insulation when a potential of 500V dc is applied between isolated pins, and between pins and the case for 10-second periods.

3.5.7 Electrical transients. The gyroscope shall withstand a spike of ±200V applied between pins A and B, C and D, or D and E for 1 millisecond.

3.5.8 Warmup. The gyroscope shall not require a warmup period of more than 60 seconds under normal operating conditions nor more than 10 minutes under the temperature extremes specified herein.

3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DOD-STD-100 shall govern the manufacturer's part number and changes thereto.

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3.7 Dimensions The outline dimensions of the gyroscope shall not exceed those shown on figure 1.

3.8 Weight. The total weight of the completely assembled gyroscope, less mounting screws and cables, shall not exceed 2.00 pounds.

3.9 Finishes and protective coatings

3.9.1 Aluminum alloy parts Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.9.1.1 Small holes and case inserts need not be anodized.

3.9.1.2 Aluminum alloys that do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

3.9.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 shall be used in lieu of anodizing.

3.9.1.4 Casting containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.9.1.5 When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.9.1.6 Parts enclosed in a hermetically sealed container need not be anodized or plated.

3.9 1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

3.9.1.8 When necessary for electrical bonding, parts need not be anodized.

3.9.2 Steel parts. Steel parts shall be coated with ion vapor deposited aluminum, where practicable, in accordance with MIL-C-83488, type I or II as applicable and of a class that is adequate to achieve the degree of protection required. Other protective coating, in lieu of MIL-C-83488, may be used if demonstrated to be satisfactory and approved by the preparing activity. Cadmium plating must be avoided when satisfactory alternative processes can be used.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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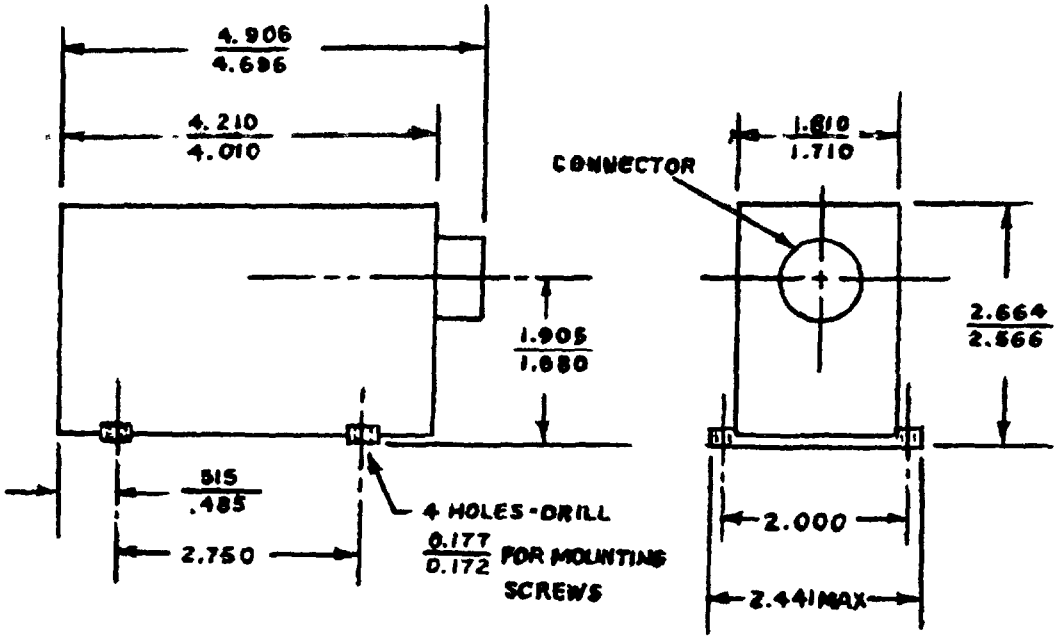


FIGURE 1. Outline and dimensions.

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3.11 Workmanship. The gyroscope, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.11.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the gyroscope, they shall be held or limited accordingly.

3.11.2 Screw assemblies. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened without damage or injury to the screw or bolt or threads.

3.11.3 Riveting. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.11.4 Cleaning. The gyroscope shall be thoroughly cleansed and loose, spattered, or excess solder, metal chips, and other foreign materials removed after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the gyroscope shall be classified as follows:

a. First article tests.....See 4.4

b. Quality conformance inspections.....See 4.5.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room

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temperature (approximately 25°C). When test are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Attitude. Unless otherwise specified, the gyroscope shall be tested in its normal operating position.

4.3.3 Voltage input. Unless otherwise specified, the gyroscope shall be tested with 115V, single-phase, 400 Hz (nominal) power.

4.4 First article testing

4.4.1 Test samples. The test samples shall consist of 12 gyroscopes representive of the production equipment. Three of the gyroscopes shall be subjected to all the tests, except reliability; and the other nine gyroscopes shall be subjected to the reliability test only (see 4.6.23). Accept-reject criteria shall be in accordance with Test Plan I of MIL-STD-781 except as modified by figure 2.

4.4.2 Test report. When the tests are conducted at a location other than the laboratory of the procurement activity, the following shall be furnished to that activity:

a. Test report - Two copies of a test report in accordance with MIL-STD-831.

b. The samples that were tested in the as-is condition.

c. Two sets of assembly drawings.

d. Brief operating instructions.

4.4.3 First article tests. The first article tests shall consist of all the tests described under 4.6.

4.5. Quality conformance inspection. The quality conformance inspection shall consist of the following:

a. Individual tests.....See 4.5.1

b. Sampling plans and tests.....See 4.5.2.

4.5.1 Individual tests. Each gyroscope shall be subjected to the following tests as described under 4.6:

a. Examination of product.....See 4.6.1

b. Power consumption test .....See 4.6.2

c. Switching rate test.....See 4.6.3

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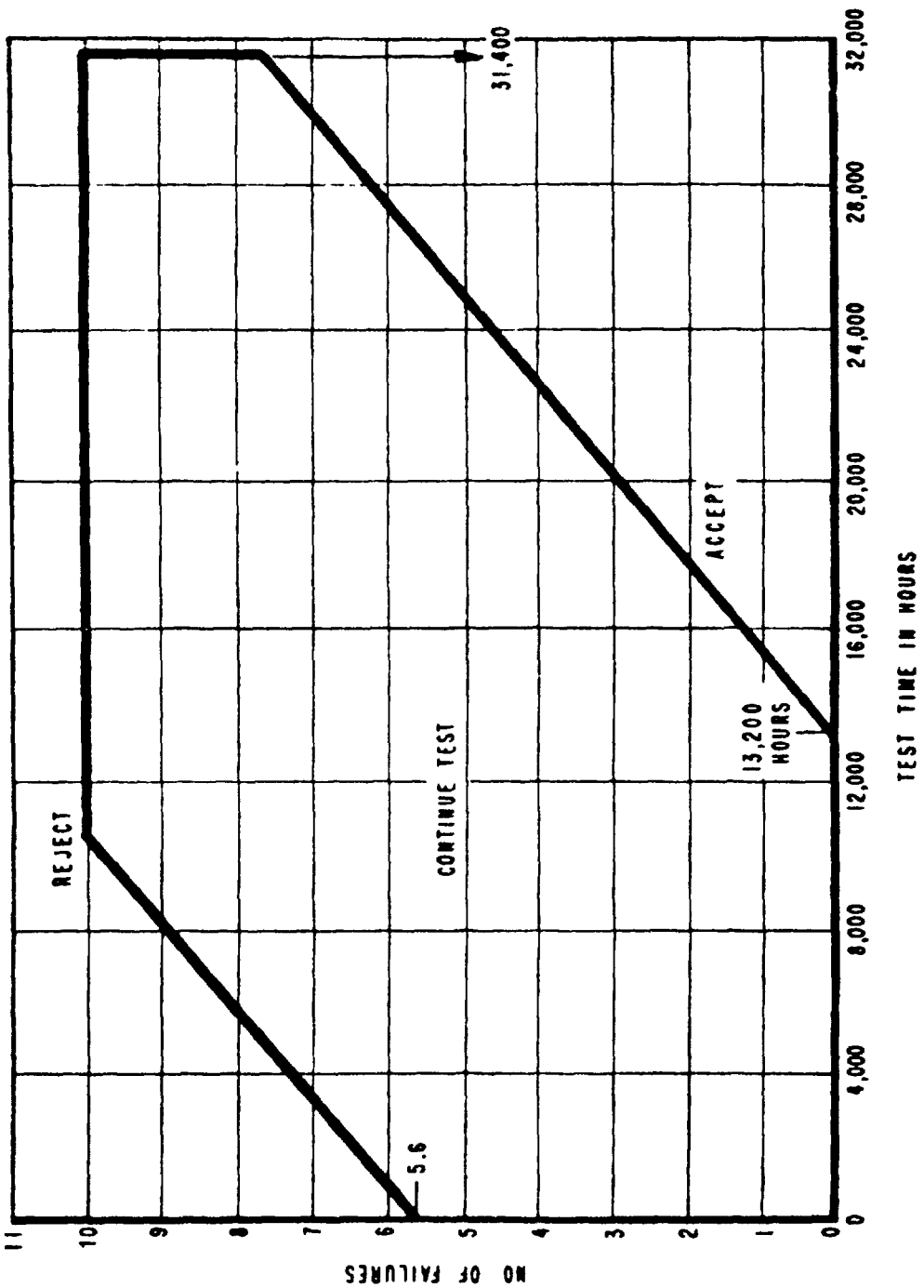


FIGURE 2. Accept continue-test reject criteria.

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d. Yaw oscillation test.....See 4.6.4

e. Steady-state turn and yaw oscillation test.....See 4.6.5

f. Power failure test.....See 4.6.6

g. Leak rate test.....See 4.6.7.

4.5.2 Sampling plans and tests

4.5.2.1 Sampling Plan A. One gyroscope selected at random from each one hundred or fraction thereof on the contract or order shall be subjected to the following tests as described under 4.6:

a. Individual tests.....See 4.5.1

b. Low temperature operation test.....See 4.6.8

c. High temperature operation test.....See 4.6.9

d. Dielectric strength test.....See 4.6.10

e. Voltage and frequency extremes test.....See 4.6.11

f. Electrical transients test.....See 4.6.12

g. Vibration error tests.....See 4.6.13

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), three gyroscopes selected at random from the first 15 items on the contract or order shall be subjected to the following tests as described under 4.6:

a. Sampling plan A tests.....See 4.5.2.1

b. Electromagnetic interference and susceptibility tests.....See 4.6.14

c. Temperature shock test.....See 4.6.15

d. Temperature-altitude test.....See 4.6.16

e. Thermal shock test.....See 4.6.17

f. Fungus test.....See 4.6.18.1

g. Salt-fog test.....See 4.6.18.2

h. Acceleration test.....See 4.6.19

i. Shock test.....See 4.6.20

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j. Vibration failure test.....See 4.6.21

k. Compatibility test.....See 4.6.22

4.5.2.3 Rejection and retest. When one gyroscope selected from a production run fails to meet the specification, items still on hand or later produced shall not be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.3.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure, but final acceptance of gyroscopes on hand or later produced shall not be made until it is determined that the items meet all the requirements of this specification.

4.5.2.4 Sampling Plan C. Unless otherwise specified (see 6.2), 7 gyroscopes shall be selected at random from the first 15 items on the contract or order and tested as specified in 4.6.23 (does not include 4.6.23.1). The accept-reject criteria shall be in accordance with test plan IV of MIL-STD-781 and shall demonstrate a minimum acceptable MTBF of 2000 hours. This testing shall include sampling plan A tests after environmental cycling. All failures which occur during sampling plan A tests shall be counted in applying the accept-reject criteria. In the event of a reject decision, production units shall be considered unacceptable and deliveries shall be stopped immediately. Deliveries shall not be made until further notice from the procuring activity.

4.5.2.4.1 Recording. Recording, data handling, and reporting procedures shall be in accordance with MIL-STD-781.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in gyroscopes already accepted. If so, the contractor shall fully advise the procuring activity of defects likely to be found and methods of correcting them.

4.5.3.1 Corrective action and retest. Corrective action and retest shall be required for each failure that can be reasonably attributed to poor manufacture or design practices, or which adversely affect mission completion.

4.6 Test methods

4.6.1 Examination of product. Prior to sealing, the gyroscope shall be inspected to determine compliance with the requirements of 3.3, 3.4.14, 3.4.15,3.6, 3.9, and 3.11. This inspection need only be performed once on each unit produced. Upon final assembly the gyroscope shall be inspected to determine compliance

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with the requirements specified herein with respect to materials, workmanship, marking, weight, and outline dimensions. The results shall be recorded. The weight and dimensions need only be recorded once for each unit.

4.6.2 Power consumption test. The total power consumption shall not exceed 10 va while operating nor 14 va while starting.

4.6.3 Switching rate test. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a rate table in its normal position and turned at a rate of 15° per minute. At this rate of turn, the internal circuit between pins D and E shall close within 30 seconds and remain closed. Likewise, the circuit between pins C and D shall open within 30 seconds and remain open. This procedure shall be repeated at 20° per minute, 45° per minute, and 180° per minute rates of turn. The same results shall occur within 20, 8, and 4 seconds, respectively. The switching rates shall be within the same limits regardless of the direction of turn. The faster rates may be tested first. This test shall be repeated with the gyroscope in the inverted position.

4.6.3.1 Return to normal. The circuit between pins D and E shall open and remain open and the circuit between pins C and D shall close and remain closed within 15 seconds after each rate of turn specified in 4.6.3 has ceased.

4.6.4 Yaw oscillation test. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a yaw test table. With zero steady-state turn imposed, the circuit between pins D and E shall remain open and the circuit between pins C and D shall remain closed when the following oscillations in yaw are imposed. This test shall be conducted with the gyroscope in the normal position and shall be repeated with the gyroscope in the inverted position.

a. ±1.0° in a 2-second period

b. ±2.0° in a 8-second period.

4.6.5 Steady-state turn and yaw oscillation test. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted in its normal position on a test apparatus capable of superimposing ±1.0° with a 2-second-period yaw oscillation on a 30° per minute rate of turn. These conditions shall be imposed on the gyroscope. The circuit between pins D and E shall close and remain closed and the circuit between pins C and D shall open and remain open within 30 seconds and shall not return to normal until the steady-state turn has ceased. The results shall be the same regardless of direction of turn. This test shall be repeated with the gyroscope mounted in the inverted position.

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4.6.6 Power failure test. The gyroscope, with power applied and operating for not less than 1 minute, shall be mounted on a rate table and turned at a rate sufficient to cause switching. After switching has occurred, power to the gyroscope shall be removed and subsequently the turn rate shall cease. The circuit between pins D and E shall open; the circuit between pins C and D shall close within 2 minutes after removal of power.

4.6.7 Leak rate test. The sealed case shall be tested for leaks by a mass-spectrometer-type helium leak detector or other means approved by the procuring activity. The leak rate shall not be sufficient to cause more than 10 percent loss or contamination of the filling medium after 1,000 hours at a differential pressure equivalent to 100,000 feet.

4.6.8 Low temperature operation. The gyroscope shall be properly connected and placed in a chamber at a temperature of -54° ±2°C for a period of 4 hours. At the end of the 4-hour period and while the gyroscope is still at the low temperature, power shall be applied. After power has been applied for 10 minutes, the gyroscope shall switch within 10, 20, and 65 seconds for rates of turn of 180°, 45°, and 15° per minute, respectively for both clockwise and counterclockwise rotation. For each rate, the switching contacts shall return to normal within 55 seconds Rotation shall alternate in this test between clockwise and counterclockwise with a 1 minute interval between each rotation to allow the gimbal time to stabilize. The faster rates may be tested first. Subsequently the gyroscope shall meet the test specified in 4.6.6 while at the low temperature. The gyroscope shall be vibrated at approximately 0.004 to 0.006 inch amplitude at a frequency of 10 to 20 Hz during this test."

4.6.9 High temperature operation test. The gyroscope shall be properly connected and placed in a chamber at a temperature of 71° ±2°C for a period of 4 hours. The gyroscope shall then be removed from the 71° C ambient and immediately placed in a 98° ±2°C chamber for a period of 10 minutes after which it shall be immediately returned to a 71°C chamber for a period of 30 minutes. The gyroscope shall be operated during this entire test. Immediately after the 30-minute soak period and while the gyroscope is still at the 71° ±2°C temperature, the switching outputs shall return to normal within 25 seconds after each turn rate is completed.

4.6.10 Dielectric strength test. A potential of 500V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.11 Voltage and frequency extremes test. The gyroscope shall be properly connected and shall be operated at normal voltage and frequency. The voltage and frequency extremes specified in 3.4.4

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shall be applied. At each of the extremes, the gyroscope shall meet the tests specified in 4.6.2 and 4.6.3. A 20-percent increase in tolerance shall be allowed for 4.6.2 and a 50 percent increase in tolerance shall be allowed for 4.6.3.

4.6.12 Electrical transients test. The gyroscope shall be subjected to an electrical spike of 200V for 1-millisecond duration in both positive and negative polarities between pins A and B, C and D, and also D and E when applying a transient to pins C and D, the gyroscope shall be operating and a rate of turn greater than 15° per minute shall be applied after which it shall meet the tests specified in 4.6.2 and 4.6.3.

4.6.13 Vibration error tests. The vibration error tests shall consist of a frequency survey with vibration applied to longitudinal, lateral, and vertical axes of the control. The control shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 Hz along the longitudinal and lateral axes of the case. The duration of the vibration shall be 1 minute in each axis for one complete cycle. The vibration test equipment employed shall not apply a yaw greater than 30° per minute to the gyroscope. No switching shall occur.

4.6.14 Electromagnetic interference and susceptibility test. The gyroscope shall pass the electromagnetic interference and susceptibility tests specified in MIL-STD-462. Test methods shall be as specified in MIL-STD-461 for class A1 equipment, except that the short duration interference requirement does not apply.

4.6.15 Temperature-shock test. The temperature shock test shall be conducted in accordance with MIL-STD-810 Method 503. After the gyroscope has returned to room temperature, the connector and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The gyroscope shall then be subjected to and shall pass the individual tests.

4.6.16 Temperature-altitude test. The gyroscope shall be subjected to a temperature-altitude test in accordance with MIL-STD-810 Method 504, class 2, procedure 1, except that the altitude shall be 80,000 feet and in step 5, the stabilized temperature shall be maintained for 4 hours. At extreme temperatures, the appropriate limits of 4.6.8 and 4.6.9 shall apply. The gyroscope shall operate without sticking. After returning to room temperature and atmospheric pressure, the gyroscope shall meet the individual tests. The gyroscope shall operate properly throughout this test and there shall be no damage that would affect subsequent operation.

4.6.17 Thermal shock test. The gyroscope case, with mechanisms removed and sealed in the same manner as the complete gyroscope,

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shall be immersed alternately in tap water maintained at 65° ±5°C and 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes with not more than 5 seconds elapsing between bath immersions. No damage to the hermetic seal shall occur as a result of this test. Following the immersions, the gyroscope case shall be subjected to and shall meet the test specified in 4.6.7.

4.6.18 Environmental test. The following environmental tests shall be conducted in accordance with the specified methods of MIL-STD-810. At the option of the procuring activity, these tests may be performed on finished empty cases with all external parts attached as on production units. At the completion of each test, the gyroscope case shall be inspected for any damage that would impair gyroscope operation.

4.6.18.1 Fungus test The fungus test shall be conducted in accordance with Method 508, procedure I.

4.6.18.2 Salt-fog test. The salt fog test shall be conducted in accordance with Method 509, procedure I, for a period of not less than 168 hours.

4.6.19 Acceleration test. The gyroscope, not operating, shall be mounted on a centrifuge in its normal operating position and accelerated, first along its vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. The gyroscope shall be subjected to an acceleration of 20g in each of the axes for 1 minute. At the conclusion of this test, the gyroscope shall be subjected to and shall meet the individual tests. No damage to the gyroscope shall result from this test.

4.6.20 Shock test. The shock test shall be conducted in accordance with MIL-STD-810, Method 516, procedures I and III. Following the test specified in procedure I, the gyroscope shall meet the individual tests.

4.6.21 Vibration failure test. The gyroscope shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514, equipment category (b), curve C, except that at 1g, the frequency shall be extended to .40 inch double amplitude and shall be maintained at this amplitude until the frequency is reduced to 5 Hz. At the completion of the vibration test, the gyroscope shall meet the individual tests.

4.6.22 Compatibility. Both output circuits shall have a maximum closed resistance of 2 ohms. Testing with 400 Hz, both output circuits shall show a minimum output impedance of 250Ω. A current of 750 milliampere shall be applied to each output circuit in turn. When appropriate in this test, the gyroscope shall be operating and a rate-of-turn greater than 15° per minute

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applied. At the completion of this test, the gyroscope shall meet the test specified in 4.6.3.

4.6.23 Reliability. The gyroscope, properly connected and with rated power applied shall be placed in a test chamber which meets the environmental-cycle conditions shown on figure 3. During the ON cycle, the gyroscope shall be exercised by a 3-axis, 30° total movement, 10 cpm, reversing scorsby action and by the vibration shown on figure 3. After each 160-hour cycle of operation, the gyroscope shall meet the individual tests. During qualification testing, an accept decision shall not be reached until the test specified in 4.6.23.1 is completed. The number of failures and hours thus accumulated shall be counted as part of this test. This testing shall include sample plan A tests after environmental cycling. All failures which occur during sampling plan A tests shall be counted in applying the accept-reject criteria.

4.6.23.1 Longevity. For first article testing, three of the gyroscopes shall be tested as specified in 4.6.23 for a total of 3,000 hours each.

4.6.23.2 Relevancy criteria. All failures shall be classified as relevant unless approved for exclusion by the procuring activity.

4.6.23.3 Test information. Sixty days prior to starting this test, the contractor shall submit detailed test information to the procuring activity for approval. This information shall include but not necessarily be limited to:

a. Equipment used in monitoring performance

b. Periodicity of calibration checks to be made on monitoring equipment

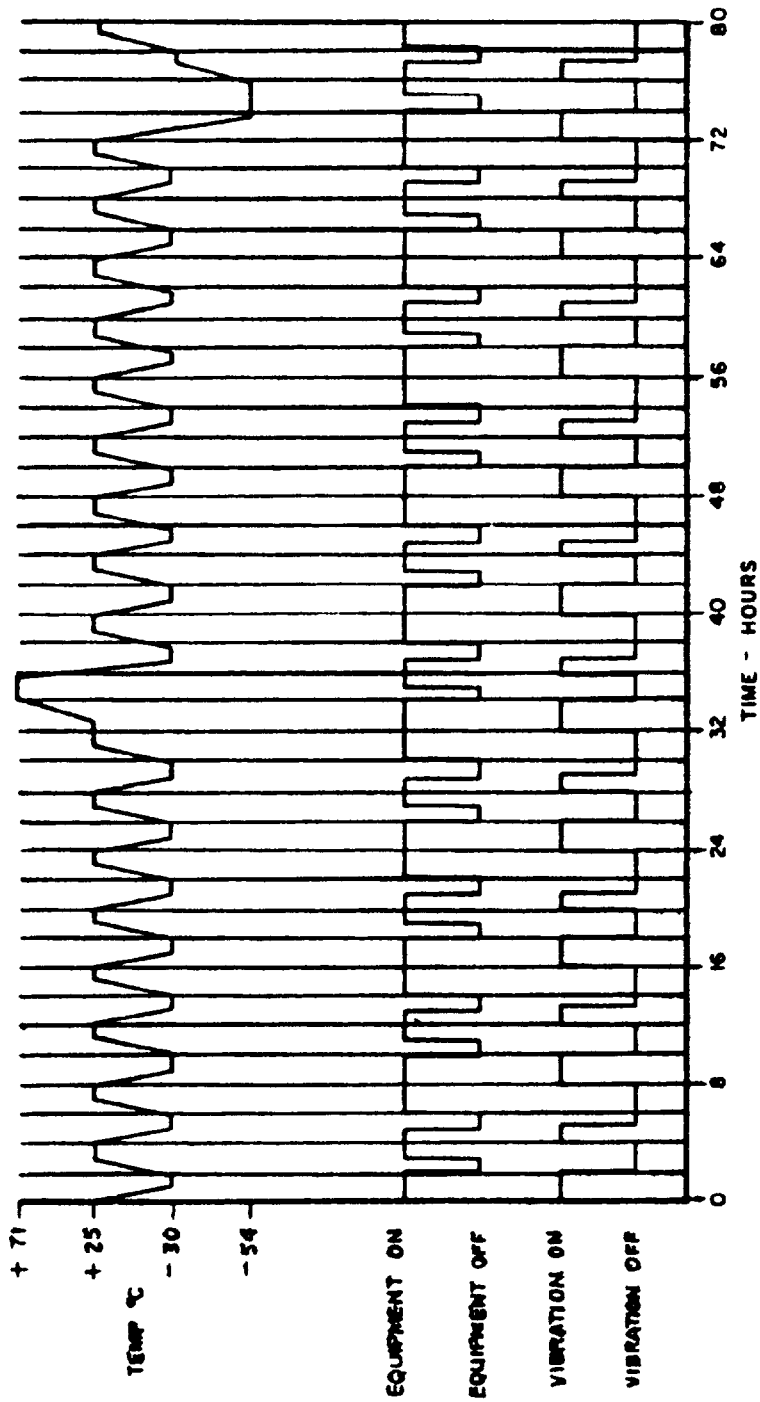
c. Description of environmental test facilities to be used, including recording facilities, methods of controlling heating and cooling facilities, and vibration equipment

d. Frequency of monitoring.

4.6.23.4 Test records. Throughout this test, the contractor shall maintain adequate test records which shall be prepared in a form suitable for reproduction. As a minimum, the records shall include the following in accordance with MIL-STD-781: An operating sheet, log failures, failure report, equipment repair sheet, failure tag, and failed-part summary analysis. Reproducible copies of the data shall be submitted to the procuring activity within 30 days after the completion of each test phase.

4.6.23.5 Repair of failed units. In the event of a failure during testing, the failed unit shall be repaired and the test continued.

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1) Temperature change 2°C/minute average. Periods of +71° and 54°C, test may be changed to coincide with working shift.
2) Equipment-off period = ½ hour for each equipment-off cycle.
3) Vibration-on period = 3 hrs/8 hrs approx as shown.
4) Temp chamber shall be stabilized at the temp shown at each hour point.
5) Environment cycle repeats at each 80 hours.
6) Vibration level - 0.5g 20 to 30 Hz.
7) Monitoring of outputs for catastrophic failure each "on" cycle.
8) Monitoring of performance conducted each 160 hours.
9) Accumulated hours = all operating time.

FIGURE 3. Reliability test environment cycles.

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4.6.24 Maintainability demonstration. The contractor shall establish the maintainability mean downtime, based on an analysis of his basic design, in accordance with MIL-STD-471, appendix B, Method 2 and shall submit his test plan to the procuring activity for approval.

4.7 Packaging inspection.

4.7.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with Group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.7.2 First article. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING.

5.1 Preservation-packaging. Preservation-packaging shall be Level A or C as specified by procuring activity.

5.1.1 Level A.

5.1.1.1 Cleaning. Unit shall be cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Drying. Immediately after cleaning, the unit shall be dried following any one of combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 Preservation application. Not required.

5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method 1A8 of MIL-P-116. Overbox each completed pack in a PPP-B-636 fiberboard, weather-resistant class box. Cushioning material between bag and containter shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4. of this specification. Insofar as practical, exterior containers shall be of uniform shape, minimum tare and cube consistent with protection required.

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5.1.2 Level C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2.d).

5.2.1 Level A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-601, boxes, wood cleated plywood. Table II of MIL-STD-794 applies for army use.

5.2.2 Level B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636 class weather-resistant.

5.2.3 Level C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 Marking. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Type MC-1 gyroscope is intended for use in aircraft to sense turns above a specified rate and switch off erection circuits of displacement gyroscopes or two-gyro platforms during these turns.

6.2 Ordering data Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests are not to be conducted

c. When sampling plan C tests are not to be conducted

d. Levels of preservation, packaging, and packing desired (see section 5)

e. All failures occurring during first article or quality conformance testing. This report will be submitted within 2 working days.

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6.3 Data. Data generated by this document is not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) referencing the appropriate data item description in the military departments' Authorized Data List (ADL). The data produced by this specification is as follows:

a. Test report in accordance with MIL-STD-831

b. Operating instructions (see 4.4.2)

c. Detailed test information (see 4.6.23.3)

d. Test records (see 4.6.23.4)

e. Report of all failures occurring during first article or quality conformance testing (to be submitted within 2 working days of failure).

6.4 Definitions. For the purpose of this specification, the following definitions will apply:

6.4.1 Failure. Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition that fails to comply with the requirements specified herein.

6.4.2 Discrimination ratio. The discrimination ratio is the ratio of the minimum acceptable MTBF to the contractually specified MTBF. In the event that any conflict in definition exists between this specification and any other document, the definition of this specification will govern.

6.4.3 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic, or both, materials. A hermetic seal is not intended to include seals accomplished by gaskets.

6.5 First article A first article is required for inspection and approval (see 3.1 and 4.4).

6.6 Marginal indicia. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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